MediLens

AI-Powered Medical Report Analyzer

Protex Hackathon 2026 — Open Innovation Track

Team MediLens

Problem Statement

Medical Illiteracy

80% of patients cannot understand their own lab reports. Complex terminology creates anxiety and confusion.

Doctor Time Constraints

Average doctor visit is only 15 minutes — not enough time to explain every metric in detail.

Language Barriers

Reports in foreign languages or technical English exclude millions from understanding their own health data.

No Trend Tracking

Patients receive isolated snapshots with no way to visualize how their health metrics change over time.

→ *1 in 5 hospital readmissions are due to patients not understanding their discharge instructions*

Proposed Solution

MediLens

An AI-powered web application that transforms complex medical reports into plain-language insights anyone can understand — with risk scoring, trend tracking, and an AI health assistant.

PDF Upload & AI Analysis

Upload any lab report PDF. AI extracts and interprets every metric.

AI Health Chat

Conversational assistant answers health questions in plain language.

Color-Coded Risk Scoring

Instant visual risk assessment: green/yellow/red for every metric.

Multi-Language Support

Understands medical reports in 40+ languages worldwide.

Health Trend Charts

Interactive Recharts dashboards track metrics across multiple reports.

Zero-Knowledge Privacy

Reports processed ephemerally. No data stored. Ever.

Technology Stack

Frontend

- React 18 + TypeScript
- Tailwind CSS + shadcn/ui
- Framer Motion (animations)
- Recharts (data viz)
- React Router (SPA routing)

Backend

- Supabase Edge Functions (Deno)
- Supabase Cloud Infrastructure
- PDF Text Extraction Engine
- RESTful API Architecture

AI / ML

- Google Gemini 3 Flash
- Lovable AI Gateway
- Structured JSON Output
- Multi-language NLP

DevOps & Security

- Lovable Cloud Deployment
- Ephemeral Processing
- CORS + Input Validation
- Zero Data Retention

Challenges Faced

PDF Text Extraction

Standard PDF libraries don't work in Deno Edge Functions. Built a custom binary parser to extract text from PDF byte streams.

✓ Custom binary PDF parser with fallback extraction

AI Response Consistency

LLM outputs varied in structure, sometimes returning markdown instead of valid JSON, breaking the frontend parser.

✓ Robust JSON cleaning with markdown stripping + fallback rendering

Medical Accuracy

Ensuring AI correctly maps lab values to reference ranges across different units and regional standards.

✓ Detailed system prompts with medical reference guidelines + disclaimer framework

Real-Time UX

AI analysis takes 3-5 seconds — needed to keep users engaged without showing a blank screen.

✓ Animated loading states with Framer Motion + optimistic UI patterns

Future Scope & Scalability

OCR for Scanned Reports

Optical character recognition to handle photo/scanned PDFs from older clinics

Persistent Health Timeline

Authenticated user accounts with cloud-stored health history and long-term trend analysis

Medication Interaction Alerts

Cross-reference medications with lab results for conflict warnings

Doctor Dashboard

Clinician-facing portal for reviewing AI summaries before patient consultations

Wearable Integration

Sync with Apple Health, Fitbit, and Google Fit for real-time vitals tracking

Enterprise API

White-label API for hospitals and telemedicine platforms at scale

Scalable Architecture: Serverless edge functions auto-scale to handle millions of concurrent analyses with zero infrastructure management

Thank You!

MediLens — Making Health Data Accessible to Everyone

Questions?